VIP – AI Hub for Config-Driven Agent Lifecycle Management, Orchestration, and Observability

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Encadrant Entreprise/ Business site Supervisor

- Nom & Prénom /Name & Surname : Ahmed Rebai

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GRADUATION PROJECT

NATIONAL ENGINEERING DEGREE

SPECIALTY : Data Science

VIP — Agentic AI Hub for Config-Driven Agent Lifecycle Management and Orchestration

By: *Ibtihel Mnaja*

Academic supervisor: *Jihen Jebri*

Corporate Internship Supervisor: *Ahmed Rebai*

VALUE

Abstract

Agentic Artificial Intelligence refers to AI systems capable of performing tasks through autonomous or semi-autonomous agents that can reason, use tools, interact with external systems, and collaborate within structured workflows. In the financial services industry, the adoption of such systems requires not only advanced AI capabilities, but also strong guarantees in terms of scalability, control, auditability, and operational reliability.

This internship report presents the design and implementation of an **AI Agentic Hub Platform** aimed at building, deploying, and operating intelligent applications at scale. The proposed platform is based on a modular multi-agent architecture in which agents are generated from declarative configurations, deployed as independent containerized services, and coordinated through an orchestration layer. This approach makes it possible to create specialized agents without modifying their source code, by relying on configurable prompts, tools, model parameters, and input/output schemas.

The work focuses on several key aspects of enterprise AI deployment, including agent generation, workflow orchestration, schema-based communication, prompt and artifact versioning using MLflow, Docker-based containerization, and observability. The objective is to provide a controlled and extensible blueprint for developing financial AI applications while reducing fragmentation, improving governance, and ensuring traceability across the full lifecycle of intelligent systems.

Keywords: Agentic AI, Multi-Agent Systems, AI Orchestration, Docker, MLflow, Financial Services, Schema Validation, Observability.

Summary

This internship project focuses on the design and implementation of **VIP – Value Intelligent Platform**, an **Enterprise-Grade Agentic AI Hub** aimed at building, deploying, and operating intelligent applications at scale.

- **Dynamic Agent Generation:** Developed a configurable Agent Factory capable of creating specialized AI agents from declarative configurations, without modifying the source code.
- **Modular and Containerized Architecture:** Designed agents as independent FastAPI services deployed in isolated Docker containers, ensuring scalability, separation of concerns, and easier maintenance.
- **Centralized Governance and Versioning:** Integrated MLflow to manage prompts, schemas, and configuration artifacts, enabling traceability, version control, and a "Zero Code Change" approach.
- **Intelligent Orchestration:** Implemented an orchestration layer supported by an adapter to coordinate multi-agent workflows and manage schema transformations between agents.

Conclusion: The platform provides a scalable and reusable blueprint for enterprise AI applications, allowing organizations to transform complex business processes into modular, auditable, and intelligent agent-based workflows.

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Glossary

ADLC:	Agent Development Lifecycle
AI:	Artificial Intelligence
API:	Application Programming Interface
IBM:	International Business Machines
LLM:	Large Language Model
QA:	Quality Assurance
REST:	Representational State Transfer
UI:	User Interface
UX:	User Experience
VIP:	Value Intelligent Platform

Introduction

The recent evolution of Artificial Intelligence has marked a major shift from traditional conversational models toward *agentic systems*. These systems are characterized by their ability to reason, use external tools, interact with data sources, and execute complex multi-step workflows with a certain level of autonomy. In this context, the subject of this internship focuses on the design and implementation of **VIP – Value Intelligent Platform**, an enterprise-grade Agentic AI Hub intended to build, deploy, and operate intelligent applications at scale across different business domains.

Despite the significant progress of Large Language Models, their integration into enterprise environments remains challenging. The main difficulty lies in transforming AI prototypes into reliable, reusable, and governable systems that can be deployed in real operational contexts. Traditional approaches often rely on hard-coded agents, making them difficult to adapt, maintain, or extend. The problem addressed in this project is therefore to design a modular platform capable of generating specialized agents from declarative configurations, while ensuring scalability, traceability, and operational control.

The adopted approach is based on a decoupled and containerized multi-agent architecture. Each agent is designed as an independent FastAPI service, deployed in a Docker container, and configured through prompts, tools, model parameters, and input/output schemas. MLflow is used to manage and version configuration artifacts such as prompts and schemas, while an orchestration layer coordinates the execution of agent workflows. A semantic adapter is also introduced to manage schema transformations between agents and ensure smooth communication across the platform.

This report is organized into five chapters detailing the progression of the work carried out during the internship. **Chapter 1** presents the general context of the internship and the host company. **Chapter 2** introduces the preliminary work completed during the onboarding phase. **Chapter 3** presents the overall architecture of the VIP platform. **Chapter 4** details the design and implementation of the agent generation mechanism and the core platform components. **Chapter 5** focuses on orchestration, observability, and evaluation aspects. Finally, a **conclusion** summarizes the main contributions of the project and proposes possible future improvements.

Chapter 1

Introduction and Project Context

1.1 Introduction

The rapid adoption of AI agents and generative language models, including large language models (LLMs) and small language models (SLMs), is creating new opportunities for enterprise automation while increasing the need for reliable orchestration, deployment, and observability. In this context, my internship at **Value Digital Services** focused on building a modular, config-driven platform for agent generation and management, alongside the evaluation of small language models through SLM-evals. This chapter introduces the host company, clarifies the project goals and problem context, and presents the methodology followed throughout the internship.

1.2 Host Company and Areas of Expertise

Value Digital Services is a Tunisian company founded in 2019 and specialized in digital transformation and AI consulting. Its activities combine software engineering, data engineering, artificial intelligence, and strategic consulting, which made it an appropriate environment for an internship focused on building an enterprise AI agent platform and evaluating small language models.

Figure 1: Logo of Value Digital Services

- **Digital Factory:** software engineers, DevOps engineers, QA engineers, business analysts, and UX/UI designers build and deliver modern digital products.
- **Data Factory:** data engineers and data analysts design data pipelines, analytics workflows, and data platforms that support decision-making.

- **AI Factory:** AI researchers and machine learning engineers design and deploy AI-driven solutions, including generative AI and intelligent automation systems.
- **Advisory:** consultants support clients in structuring digital transformation initiatives, aligning technical choices with business and governance needs.

These poles work together to deliver modern digital solutions, from consulting and design to data infrastructure, AI implementation, deployment, and operational support.

1.3 Project Overview

The project, entitled **VIP – Value Intelligent Platform**, aims to design and implement an enterprise-grade AI agent orchestration platform capable of building, deploying, and managing intelligent applications through a modular multi-agent architecture. Unlike traditional monolithic AI systems, the platform follows an *Agent-as-a-Service* approach, where specialized agents are dynamically generated from declarative configurations and deployed as independent containerized services.

VIP provides a unified environment for agent creation, workflow orchestration, observability, and lifecycle management. Users can configure agents by selecting models, prompts, schemas, providers, and tools without directly modifying the source code. The generated agents can then be integrated into multi-agent execution pipelines coordinated through an orchestration layer and monitored through built-in observability services.

This platform directly addresses the gaps identified in existing solutions by combining support for multiple LLM providers and open-source models with a modular architecture, declarative configuration management, and built-in observability. This design reduces dependency on a single proprietary provider, simplifies agent creation and deployment, and makes agent behavior easier to trace, evaluate, and improve over time.

Alongside VIP, I developed the **SLM-evals** side project, a reproducible evaluation pipeline for benchmarking LLM and SLM providers. It compares OpenAI, Groq, Phi-4, Llama 3.1, and other alternatives under controlled prompts and golden datasets, while tracking latency, token usage, and layered evaluation metrics. The results of this pipeline guide the choice of models for the *LLM-provider* layer and support VIP's objective of maintaining provider flexibility while selecting models that fit performance, cost, and reliability requirements.

1.4 Problem Statement

Despite the rapid evolution of AI agents and large language models, the deployment of agentic systems in enterprise environments remains complex. These challenges matter because enterprises need systems that can scale reliably, remain auditable, control operational costs, and comply with internal governance and security requirements. When agent platforms lack provider flexibility, observability, automation, or modular orchestration, they become difficult to deploy at scale and hard to maintain in production.

The main challenges addressed by this project can be summarized as follows:

- **High Operational Costs and Vendor Lock-in:** Existing AI agent solutions can be expensive to operate at scale, especially when they depend heavily on proprietary LLM APIs. This dependence increases costs, limits provider flexibility, and makes it harder to switch between OpenAI, Groq, Azure, open-source LLMs, or SLM alternatives.
- **Poor Observability:** Many current systems lack sufficient traceability, monitoring, and execution tracking. This limits debugging, performance evaluation, and the audit evidence required for compliance and governance.
- **Absence of Automated Deployment:** Creating and deploying new agents is often a manual and time-consuming process. Without automated deployment mechanisms, it becomes difficult to rapidly generate, configure, and launch agents in a consistent way.
- **Lack of Orchestration and Modularity:** Existing approaches often do not provide strong workflow orchestration capabilities. Agents are not always designed as reusable and modular components, which limits workflow composition and makes multi-agent collaboration harder to manage.

Based on these limitations, the central problem of this project can be formulated as follows:

> How can we design a modular, secure, observable, and cost-optimized AI agent orchestration platform that enables dynamic agent creation, automated deployment, and workflow composition while evaluating and integrating multiple LLM and SLM providers to reduce vendor lock-in, control costs, and maintain strong performance?

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1.5 Objectives

Based on the problem statement, the objectives of the project are divided into functional and non-functional objectives.

Functional Objectives

- Enable dynamic agent creation from declarative YAML or JSON configuration files.
- Provide a visual workflow builder for composing and managing multi-agent execution pipelines.
- Offer a unified interface for selecting models, prompts, schemas, providers, and tools without modifying the source code.
- Support reusable multi-agent pipelines that can combine specialized agents into structured workflows.
- Integrate SLM-evals results into the provider selection process to compare LLM and SLM options before deployment.

Non-functional Objectives

- Reduce vendor lock-in and operational cost by evaluating multiple providers, including proprietary LLM APIs and local SLM alternatives.
- Optimize deployment costs through containerized, configurable, and reusable agent services.
- Ensure observability and traceability by monitoring execution traces, token usage, latency, and estimated cost.
- Preserve modularity and scalability by allowing agents, tools, providers, and workflows to evolve independently.
- Strengthen security and provider independence by limiting hard-coded provider dependencies and supporting controlled configuration management.

1.6 Technology Choices

The project relies on a modular technology stack selected to support agent generation, orchestration, deployment, observability, and provider flexibility.

1.6.1 Development Frameworks and Dev Environment

- **FastAPI:** lightweight Python framework used to build high-performance asynchronous APIs for agents and platform services.

Figure 2: Logo of FastAPI

- **Next.js:** React-based framework used to build the web frontend and provide a modular interface for agents, workflows, and configurations.

Figure 3: Logo of Next.js

- **LangChain:** library used to construct LLM-driven agents, connect tools, and manage prompt-based application logic.

Figure 4: Logo of LangChain

- **LangGraph:** library used to define graph-based orchestration logic for stateful multi-agent workflows.

Figure 5: Logo of LangGraph

- **Jupyter Notebook:** interactive environment used during prototyping, experimentation, SLM-evals, and prompt engineering.

Figure 6: Logo of Jupyter Notebook

1.6.2 LLM and SLM Providers

- **OpenAI:** commercial LLM provider used to access GPT-series models for high-quality generation and evaluation tasks.

Figure 7: Logo of OpenAI

- **Groq:** commercial inference provider used to access fast LLM serving, including Llama-based models.

Figure 8: Logo of Groq

- **Microsoft Phi-4:** small language model considered for cost-efficient and potentially self-hosted inference.

Figure 9: Logo of Microsoft Phi-4

- **Meta Llama :** open-weight LLM used as an alternative provider option for flexible deployment and benchmarking.

Figure 10: Logo of Meta Llama

- **SLM-evals:** used to benchmark providers and models in order to choose the best mix for the VIP provider layer.

1.6.3 Database, Storage, and Experiment Tracking

- **PostgreSQL:** relational database used to store configurations, metadata, workflows, and other structured platform data.

Figure 11: Logo of PostgreSQL

- **MLflow:** experiment tracking and artifact registry used to version models, prompts, schemas, datasets, and evaluation metrics.

Figure 12: Logo of MLflow

1.6.4 Deployment and Containerization

- **Docker:** container platform used to package individual agents, the builder, the orchestrator, and supporting services into reproducible images.

Figure 13: Logo of Docker

- **Docker Compose:** used during development and testing to run multi-service setups with consistent networking and configuration.

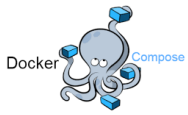

Figure 14: Logo of Docker Compose

1.6.5 Observability and Monitoring

- **Laminar:** used to capture execution traces, token usage, latency, and cost metrics across agents.

Figure 15: Logo of Laminar

1.7 Methodology

1.7.1 Methodological Approaches Considered

Several methodological approaches were considered in order to structure the development of the VIP platform. The project combines software engineering, data science, agent lifecycle management, orchestration, deployment, and observability. Therefore, the selected methodology had to support both iterative development and long-term architectural control. The approaches considered were Scrum, IBM Data Science Methodology, and IBM Agent Development Lifecycle (ADLC).

1.7.2 Scrum Agile Methodology

Scrum is an Agile framework based on iterative and incremental development. The official Scrum Guide describes Scrum as an iterative and incremental approach used to optimize predictability and control risk. It supports progressive delivery by organizing work into short cycles, reviewing progress regularly, and adapting the next development steps according to feedback.

Scrum was considered because the project required frequent adjustments, progressive development, and regular validation of features such as agent creation, workflow orchestration, prompt management, and observability dashboards.

However, Scrum alone was not sufficient for this internship because the project required more than sprint-based feature delivery. It also required a structured lifecycle for AI/data-related work, architectural traceability, governance, deployment control, and continuous monitoring of AI agents. For this reason, Scrum-inspired practices were retained for implementation organization, but Scrum was not selected as the main project methodology.

1.7.3 IBM Data Science Methodology – The Foundational Methodology

The IBM Data Science Methodology was considered as the foundational methodology because it provides the macro lifecycle of the project. The CognitiveClass data science methodol-

ogy is structured around phases that move from business understanding and analytic approach to data requirements, collection, understanding, preparation, modeling, evaluation, deployment, and feedback. This structure makes it suitable for organizing an AI-oriented project from problem definition to operational feedback.

The methodology consists of 10 stages organized in a circular workflow:

- **Business Understanding:** defining the problem to be solved and establishing the main objectives.
- **Analytic Approach:** determining the appropriate analytical and technical approach to address the problem.
- **Data Requirements:** identifying the information and resources required to address the problem.
- **Data Collection:** gathering the required data and project inputs from the relevant sources.
- **Data Understanding:** exploring and examining the available information in order to understand quality, constraints, and relationships.
- **Data Preparation:** cleaning, transforming, and formatting the required data or resources for later use.
- **Modeling:** building predictive, descriptive, or intelligent models using appropriate algorithms and implementation choices.
- **Evaluation:** assessing the obtained results against the initial objectives and expected quality criteria.
- **Deployment:** implementing the solution in a production or operational environment.
- **Feedback:** monitoring performance, collecting feedback, and iterating on earlier stages when improvements are required.

Figure 16 presents the original IBM Data Science Methodology lifecycle, which was used as the main macro-framework for structuring the project from business understanding to deployment and feedback.

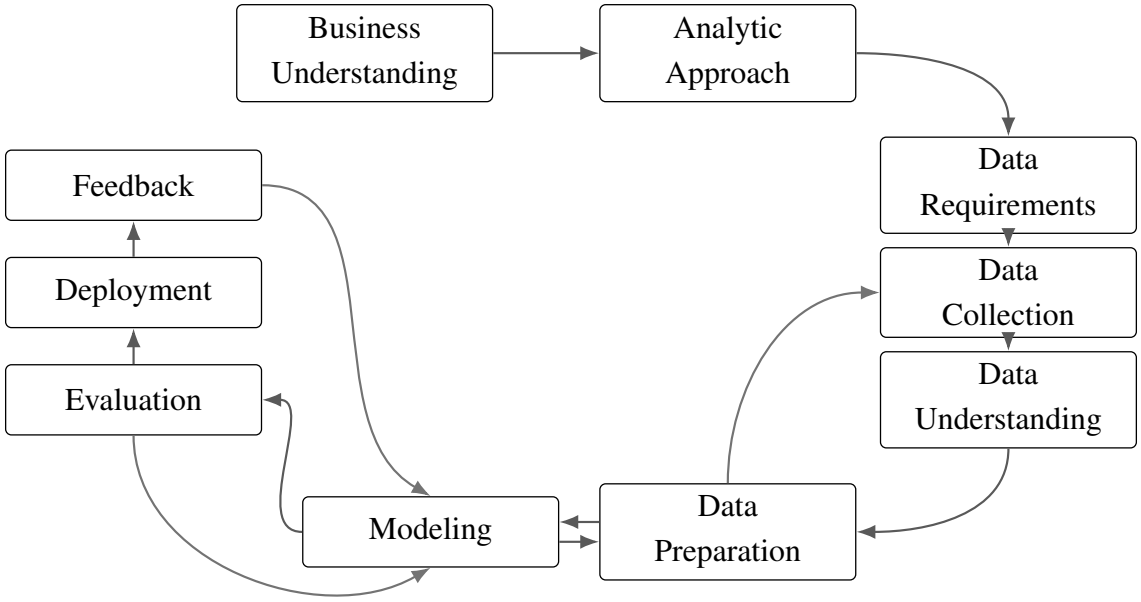

Figure 16: Original IBM Data Science Methodology lifecycle

This methodology is relevant for the VIP project because it supports a structured understanding of the business problem, the identification of requirements, the preparation of resources, the evaluation of outputs, and the deployment of the final solution. It also provides a feedback loop, which is important for improving agent behavior, model selection, and workflow reliability over time.

1.7.4 IBM Agent Development Lifecycle

While the IBM Data Science Methodology provides the overall project structure, it does not fully describe the specific lifecycle of AI agents. The VIP platform required a more agent-specific perspective because agents are generated from configuration, connected to tools, deployed as services, monitored, and improved after execution. For this reason, ADLC was used as a complementary framework.

IBM defines ADLC as a structured lifecycle for building, deploying, optimizing, and managing AI agents through distinct phases and iterative loops. In the context of VIP, this perspective is relevant because agents must remain reliable, observable, auditable, and aligned with business requirements after deployment.

In this report, the ADLC perspective is summarized through the following phases:

- **Plan:** alignment of stakeholders, use cases, objectives, success metrics, expected agent behavior, and operating procedures.
- **Code & Build:** implementation of agents through model selection, prompt design, tool integration, orchestration logic, version control, and controlled development environments.
- **Test & Release:** evaluation of agents against predefined criteria, including policy checks, security validation, and release readiness before cataloging or production use.

- **Deploy:** controlled rollout of validated agents into production environments with governance, versioning, rollback, and runtime security mechanisms.
- **Operate:** continuous observation and optimization of deployed agents using operational metrics such as accuracy, latency, cost, and user satisfaction.
- **Monitor:** ongoing auditing of agents and tools to support fairness, transparency, compliance, observability, and reproducibility.

Figure 17 illustrates the IBM Agent Development Lifecycle, which was used to guide the agent-specific phases of the project.

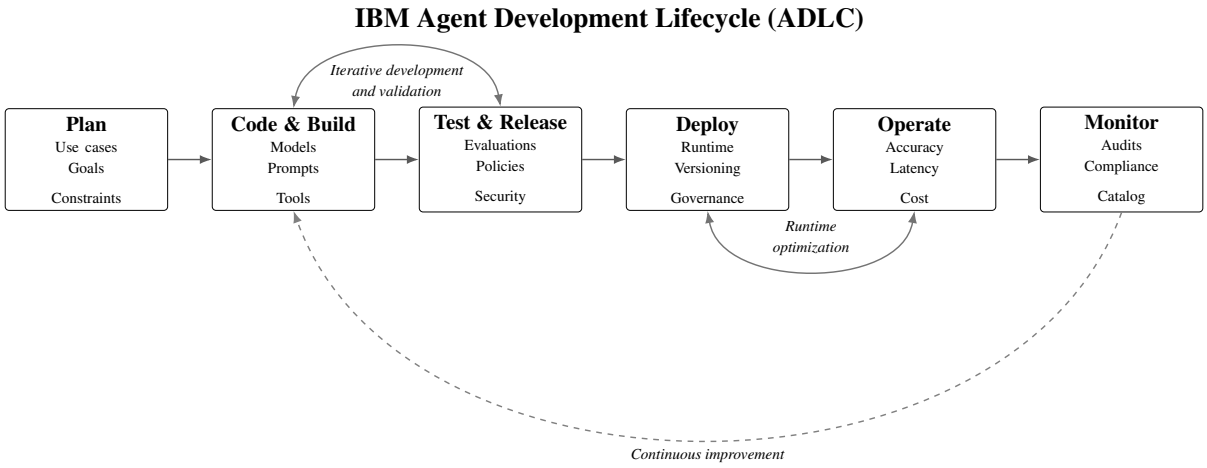

Figure 17: IBM Agent Development Lifecycle (ADLC)

In the VIP project, the Plan phase corresponds to defining agent use cases, expected behavior, schemas, and constraints. The Code & Build phase corresponds to configuring prompts, tools, models, and runtime artifacts. The Test & Release phase corresponds to validating agent outputs and schema compliance. The Deploy phase corresponds to packaging agents as Dockerized FastAPI services. The Operate and Monitor phases correspond to observing latency, token usage, cost, and execution traces.

IBM also describes agent building as the transformation of intent and design into concrete capabilities by configuring models, behavior, knowledge, and tools. This is aligned with the VIP approach, where agents are created from declarative configurations including prompts, schemas, tools, and model parameters.

1.7.5 Comparative Analysis of the Methodologies

The three methodologies were compared according to their relevance to the VIP project, especially with regard to flexibility, strategic vision, AI project support, agent lifecycle coverage, deployment, monitoring, and limitations.

Criterion	Scrum	IBM Data Science Methodology	IBM ADLC
Approach	Iterative and sprint-based	Structured and data/AI-oriented	Agent lifecycle-oriented
Flexibility	High	Medium to high	High
Strategic vision	Limited to product increments	Strong	Focused on agent operations
Fit for AI projects	Partial	Strong	Strong for agentic AI
Fit for agent lifecycle	Limited	Partial	Very strong
Deployment and monitoring support	Not detailed	Present through deployment and feedback	Central through deploy, operate, and monitor
Main limitation	Not enough governance for AI agents	Not specific enough for agent runtime behavior	Too agent-specific to structure the whole project alone

Table 1: Comparison of the methodological approaches considered for the VIP project

The comparison shows that none of the methodologies fully covers the needs of the VIP project when used alone. Scrum supports practical development organization, IBM Data Science Methodology provides the strategic project lifecycle, and ADLC addresses the agent-specific lifecycle required by configurable, deployable, and observable AI agents.

1.7.6 Adopted Hybrid Methodology for the VIP Project

Based on the comparison, the adopted methodology is a hybrid approach combining IBM Data Science Methodology and IBM ADLC, while keeping Scrum-inspired iterations for practical development organization.

IBM Data Science Methodology was selected as the main macro-framework because it structures the project from business understanding to deployment and feedback. It ensures that the project remains aligned with business objectives, technical constraints, and evaluation criteria.

ADLC was added because the VIP platform is centered on AI agents. The project required agent-specific phases such as planning agent behavior, configuring prompts and tools, validating outputs, deploying agents as services, operating them, and monitoring their execution.

Scrum was not selected as the main methodology because the project required stronger architectural governance and lifecycle control than a sprint-based framework alone could provide. However, Scrum-inspired practices were useful during implementation, especially for breaking the work into incremental tasks, validating features progressively, and adapting to feedback.

This hybrid methodology was therefore chosen because it provides a balance between structure, flexibility, and agent-specific lifecycle management. It supports the strategic organization of the project, the iterative improvement of AI components, and the operational requirements of an enterprise-grade agentic platform.

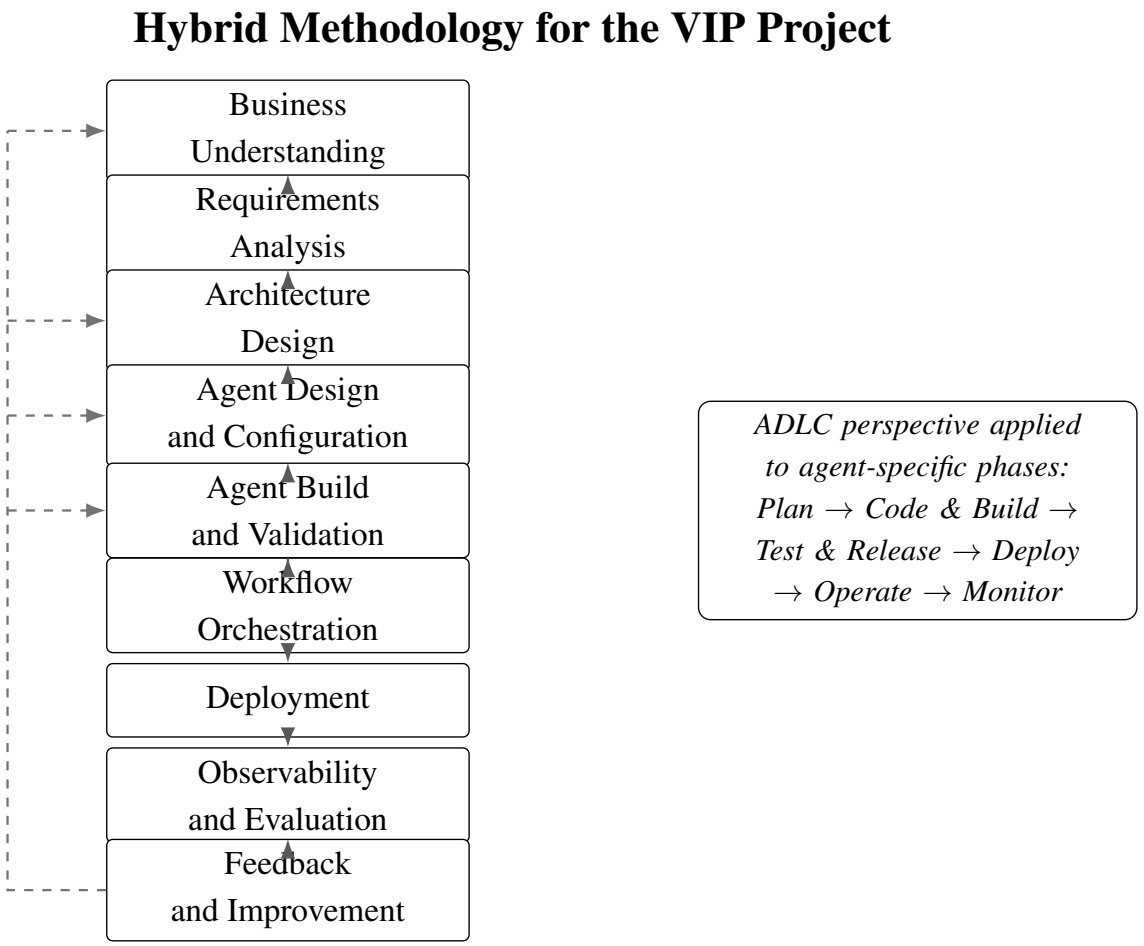

Figure 18: Hybrid methodology adopted for the VIP project

1.8 Project Timeline

The project officially started on December 1, 2025 and is scheduled to end on May 31, 2026, spanning a duration of six months. This timeframe was organized to support a structured and progressive approach, covering onboarding, research, design, development, integration, observability, and testing phases.

1.9 Summary

This chapter introduced the context of the internship by presenting the motivation behind enterprise AI agent orchestration, the host company Value Digital Services, and the objectives of the VIP project. It also clarified the main problem addressed by the project, namely the need for a modular, observable, cost-optimized, and provider-independent platform for creating and managing AI agents. The chapter then summarized the selected technologies, including FastAPI, Next.js, LangChain, LangGraph, Docker, MLflow, PostgreSQL, Laminar, and the LLM/SLM providers evaluated through SLM-evals, before presenting the methodological approach and the project timeline.

The next chapter focuses on the state of the art. It reviews existing agent frameworks, LLM providers, orchestration approaches, and evaluation methods in order to position the proposed

solution with respect to current tools and practices.

Chapter 2

State of the Art

2.1 State of the Art of AI Agents and Orchestration

2.1.1 Defining AI Agents

An AI agent is a software entity capable of perceiving inputs, reasoning over context, invoking external tools, and producing structured outputs in order to achieve a defined objective. Unlike traditional prompt-response systems, agentic systems extend large language models (LLMs) with execution logic, memory, tool interfaces, and multi-step reasoning capabilities.

In enterprise environments, an agent is not merely a conversational component, but an operational unit that interacts with APIs, databases, and external systems. It may retrieve information, transform data, validate outputs, or trigger automated workflows. As a result, agents must be treated as managed execution components rather than isolated prompt wrappers.

This evolution from simple generation models to operational agents introduces new requirements: lifecycle management, configuration control, structured outputs, monitoring, and governance.

2.1.2 Single-Agent and Multi-Agent Systems

A single-agent system relies on one autonomous unit responsible for handling the entire task loop, including reasoning, tool invocation, and response generation. This approach is suitable for bounded or narrowly scoped tasks.

However, complex enterprise workflows often require decomposition into multiple responsibilities such as planning, retrieval, validation, transformation, and reporting. Multi-agent systems distribute these responsibilities across specialized agents, each with a defined role and structured input/output schema.

In such architectures, agents collaborate through intermediate outputs and shared state. While this improves modularity and reuse, it significantly increases coordination complexity. Consequently, multi-agent systems require orchestration mechanisms to manage sequencing, state transitions, schema compatibility, and error handling.

2.1.3 Why Orchestration Is Required

As agentic systems evolve from isolated assistants to collaborative execution networks, orchestration becomes a fundamental architectural requirement. Orchestration ensures that agents are executed in the correct order, that data flows between them in a structured manner, and that

failures can be handled predictably.

Without orchestration, agent logic becomes embedded in ad hoc control code, making workflows difficult to reproduce, debug, or scale. Orchestration frameworks introduce structured patterns such as workflow graphs, state machines, retries, and task delegation.

In enterprise contexts, orchestration is not only about execution sequencing. It also ensures traceability, auditability, and governance of agent interactions.

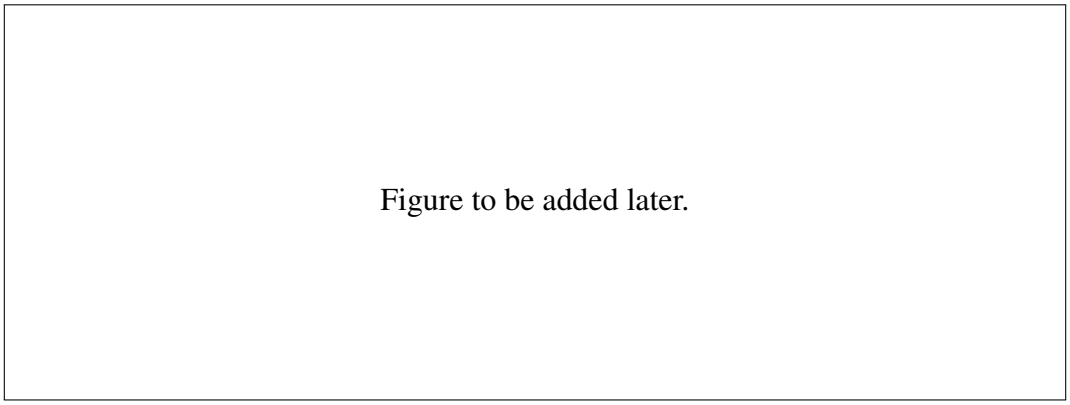

Figure 19: Evolution from single-agent execution to orchestrated multi-agent workflows.

2.1.4 Provider Landscape: Frontier LLMs and Small Language Models

The provider layer is a strategic dimension of agentic systems. Frontier LLMs such as GPT-series or other commercial APIs offer strong reasoning performance and ease of integration. However, they introduce dependency on external infrastructure, token-based operational costs, and potential vendor lock-in.

Small Language Models (SLMs) and open-weight models offer an alternative path. They may be self-hosted or deployed closer to the application environment, improving latency control and governance. However, their suitability depends on rigorous evaluation against task-specific requirements.

For enterprise platforms, provider flexibility becomes essential. A robust architecture should abstract the provider layer, enabling model replacement based on cost, latency, compliance, or performance constraints.

2.2 Preliminary Study and Project Specifications of the VIP Project

2.2.1 Problem Statement

Despite the rapid evolution of agentic AI frameworks and language models, enterprise deployment remains fragmented. Existing solutions often provide partial capabilities, focusing either on workflow automation, agent collaboration, or managed cloud execution.

However, organizations require platforms that combine:

- Configuration-driven agent generation
- Provider flexibility

- Workflow orchestration
- Deployment automation
- Observability and governance

The central challenge addressed by VIP is therefore:

> How can we design a modular, observable, provider-neutral AI agent orchestration platform capable of generating and managing agents dynamically while maintaining cost control, scalability, and governance?

2.2.2 Business Objectives

The business objectives of VIP are:

- Reduce vendor lock-in by abstracting model providers
- Enable rapid agent creation without modifying source code
- Improve governance through versioning and traceability
- Reduce operational cost through modular deployment
- Provide reusable workflows for domain-specific AI applications

2.2.3 Technical Objectives

From a technical perspective, the VIP project aims to provide a modular platform for creating, deploying, orchestrating, and supervising AI agents. The platform must support configurable agent behavior, reusable workflow composition, provider abstraction, and structured communication between agents.

These objectives are not limited to model performance. They also concern the way agents are generated, connected, monitored, and maintained throughout their lifecycle. Therefore, the main technical objectives of VIP are to enable configuration-driven agent creation, standardize agent communication through schemas, support multiple LLM and SLM providers, and ensure traceability during execution.

The model evaluation aspect is supported by the SLM-evals second project, which is presented separately in Section 2.6.

The main technical objectives of VIP are:

- enable dynamic agent creation from declarative configuration files;
- support reusable multi-agent workflows;
- provide a unified provider abstraction layer;
- ensure structured input and output validation through schemas;
- support prompt and schema versioning;
- provide observability over agent execution, token usage, latency, and errors;
- allow future integration of model evaluation results into provider selection.

2.2.4 Functional and Non-Functional Requirements

2.2.4.1 Functional Requirements

- Dynamic agent creation from YAML/JSON configuration files
- Model, provider, temperature, and prompt selection
- Tool binding through a tools catalog
- Structured schema-based input and output validation
- Multi-agent workflow composition
- Visual dashboard for managing agents and workflows
- Versioning of prompts and schemas using MLflow

2.2.4.2 Non-Functional Requirements

- Scalability through containerized deployment
- Performance optimization via lightweight Docker images
- Observability of API calls, token usage, and latency
- Reliability through schema validation and structured outputs
- Maintainability through versioned configuration artifacts
- Provider independence

2.2.5 Success Criteria

VIP is considered successful if:

- Agents can be generated and deployed without modifying source code
- Multi-agent workflows execute reliably
- Execution traces are observable
- Provider switching can be performed without architectural modification
- Evaluation results guide provider selection decisions

2.3 Comparison of Existing Platforms with Respect to VIP

2.3.1 Evaluation Criteria Derived from VIP Requirements

The comparative study of existing platforms is derived directly from the needs of VIP rather than from generic feature lists. The first criterion is the orchestration model, because VIP must support both simple flows and more advanced multi-agent coordination patterns. The second criterion is provider flexibility, since the platform must avoid hard dependency on a single LLM vendor and must remain open to commercial APIs as well as smaller or open models. The third criterion is observability, which includes execution traces, latency visibility, token and cost tracking, and debugging support. The fourth criterion is the deployment model, because VIP targets reproducible packaging and controlled execution environments rather than ad hoc local scripts only. The fifth criterion is scalability, meaning the ability to support increasing workflow load, agent concurrency, and operational growth. The sixth criterion is ease of use, since one of the platform goals is to reduce the friction of creating and operating agents. The last criterion is enterprise readiness, which covers access control, configuration management, security integration, and operational governance. These criteria are consistent with current documentation for workflow platforms, agent frameworks, and observability tools, and they also align with the broader recommendation in evaluation literature to assess LLM systems with multiple dimensions rather than accuracy alone.

2.3.2 Comparative Study of n8n, CrewAI, and Jarvio

n8n is best understood as a workflow automation platform with AI extensions. Its AI Agent node is tool-oriented and explicitly requires at least one connected tool sub-node, while its LangChain cluster node structure shows that AI execution is embedded inside a broader node-based workflow model. n8n also documents scalable execution through queue mode, where workflow execution is delegated from a main instance to worker instances through Redis, and it exposes OpenTelemetry tracing for workflow and node executions in self-hosted deployments. This makes n8n attractive for visual orchestration, fast integration, and practical automation. However, its abstraction remains workflow-centric rather than agent-platform-centric: it does not natively target configuration-driven generation of reusable agent services with an internal provider benchmarking loop.

CrewAI is stronger on the agentic side. Its documentation defines Crews as collaborative groups of agents and Flows as structured, event-driven workflows that manage state and control execution. It also supports sequential and hierarchical processes, where a manager model or manager agent can coordinate delegation and validate outcomes. In addition, CrewAI supports multiple LLM providers, YAML-based configuration, direct code-based model configuration, and observability integrations including MLflow. This makes CrewAI a stronger reference for planner-worker collaboration, controlled multi-agent execution, and provider abstraction. Its main limitation with respect to VIP is that it remains primarily a development framework: it helps build agent systems, but it does not by itself provide the full managed platform layer that VIP requires for declarative agent creation, standardized container packaging, visual lifecycle management, and integrated provider evaluation.

Jarvio is considered here as a managed operational platform candidate rather than as a code-first orchestration framework. In practical terms, it is most useful as a contrast case: it represents the category of managed AI business platforms that can reduce infrastructure burden and accelerate operational adoption. However, during this review, public engineering documentation for Jarvio was not available at the same level of detail as the documentation for n8n and CrewAI. For this reason, the comparison remains conservative: Jarvio can be positioned as easier to adopt operationally, but less transparent from an architectural perspective, less portable than a self-hosted stack, and less suitable as a technical foundation when the objective is to control provider abstraction, deployment topology, schema governance, and runtime traces at platform level. This limited public technical transparency is itself a relevant gap for VIP, whose target is an enterprise-grade platform with explicit control over architecture and operations.

Table 2: Comparative study of n8n, CrewAI, Jarvio, and VIP implications

Criterion	**n8n**	**CrewAI**	**Jarvio**	**VIP implication**
Orchestration model	Visual node-based workflows with AI agent nodes	Crews for collaboration, Flows for structured execution	Managed agent/workflow experience at product level	VIP should combine visual orchestration with explicit multi-agent control
Provider flexibility	Moderate; AI integrations are broad, but model strategy is not the core design axis	High; supports multiple LLM providers and configurable model bindings	Not sufficiently documented publicly at engineering depth	VIP must keep provider abstraction explicit and replaceable
Observability	Good in self-hosted mode through executions, metrics, and OpenTelemetry tracing	Good through tracing and multiple observability integrations	Not sufficiently documented publicly at engineering depth	VIP must treat tracing, cost, and latency as built-in capabilities
Deployment model	Self-hosted and cloud; scalable queue mode with workers	Primarily framework-oriented; deployable but requires surrounding platform choices	Managed platform orientation	VIP should provide standardized containerized deployment units
Scalability	Strong for workflow execution through queue mode and workers	Good at application level, but depends on surrounding infrastructure choices	Operational scaling is product-facing, not deeply documented technically	VIP should isolate orchestration scaling from model/provider dependence
Ease of use	High for workflow composition and integrations	Moderate; more code-oriented and developer-centric	Likely high from a business-user perspective	VIP should reduce setup complexity without losing architectural control
Enterprise features	RBAC, custom roles, SAML/OIDC, source control, log streaming, insights	Enterprise-oriented positioning plus observability and security emphasis	Product-level operational positioning	VIP should expose governance and operations without vendor lock-in

The table above is derived from n8n's AI Agent, queue mode, OpenTelemetry, and enterprise documentation, and from CrewAI's Crews, Flows, provider, and observability documentation. The Jarvio row is intentionally cautious because comparable public engineering documentation was not available during this review.

2.3.3 Discussion of Platform-Level Gaps

The comparison shows that the three platforms address different parts of the problem space, but none of them maps exactly to the full VIP requirement set. n8n is excellent for visual automation and systems integration, yet its core abstraction remains the workflow rather than the lifecycle-managed agent service. CrewAI is much closer to VIP on agent design and orchestration logic, but it still assumes a developer-centric framework context and leaves significant platform concerns to surrounding infrastructure choices. Jarvio, by contrast, is valuable as an example of managed operational packaging, but its public technical depth is insufficient for a platform whose architectural decisions must be justified in an engineering report. Therefore, the gap is not that such tools are weak; rather, each tool covers one slice of the lifecycle, while VIP aims to combine agent definition, provider abstraction, orchestration, observability, and evaluation in one controlled architecture.

2.4 Evaluation and Operational Foundations

2.4.1 Evaluation Frameworks for Agentic Systems

Evaluation in agentic systems cannot be reduced to a single leaderboard score. The literature on language-model evaluation recommends multi-dimensional assessment, and this is even more important for agent workflows where errors may arise from reasoning, tool use, retrieval, formatting, or coordination rather than from generation quality alone. A robust evaluation framework therefore combines logic-based checks, reference-based metrics when available, model-based judging for open-ended outputs, and operational measurements such as latency and cost. Laminar's evaluation model is consistent with this view: it distinguishes executors, evaluators, datasets, and result visualization, and explicitly supports both logic-based evaluation and model-based evaluation. In the same direction, MLflow positions tracing and evaluation as complementary, so that quality analysis can be inspected together with internal execution data. In research, HELM argues for multi-metric evaluation beyond accuracy, G-Eval shows the usefulness of LLM-as-a-judge for open-ended quality assessment, and frameworks such as RAGAS and ARES show that faithfulness and relevance must be assessed separately in retrieval-grounded systems.

2.4.2 Observability and Governance Requirements

For VIP, observability is not an optional monitoring add-on but a structural requirement. A platform that creates and orchestrates AI agents must expose what the agent received, what model or tool it called, how long each step took, how many tokens were consumed, and where failures or invalid transitions occurred. Laminar is designed precisely for this space: it traces LLM calls, tool use, and custom functions; supports evaluations, datasets, alerts, and dash-

boards; and can run in managed cloud or in a self-hosted Docker Compose deployment. MLflow complements this by covering prompt lifecycle and traceability: its tracing captures inputs, outputs, and metadata for intermediate steps, while its Prompt Registry provides immutable prompt versions, side-by-side diffs, and integration with tracing and evaluation. Governance also includes access control and environments. On the workflow-platform side, n8n documents RBAC, custom roles, SAML/OIDC, source control, and log streaming, which confirms that enterprise AI systems require operational governance in addition to workflow logic. These elements justify why VIP must be designed around traceability, configuration history, and controlled operations from the start.

2.4.3 Why Observability Is a First-Class Need for VIP

VIP targets multi-step, multi-agent, provider-flexible execution. In such a setting, the final answer alone is not enough to diagnose system behavior. Failures may come from the wrong prompt version, a schema mismatch between agents, a tool error, excessive latency on one provider, or unstable output formatting. Laminar's trace model illustrates why deep observability matters: it captures LLM calls, tool calls, and custom functions, supports replay and evaluation, shows token and cost badges in multi-agent traces, and provides trace-level analysis and alerts. MLflow adds the missing governance layer around prompt evolution by making prompt versions immutable and connecting prompt changes to tracing and evaluation results. In the VIP context, this is why the observability layer must be introduced already in Chapter 2 as a design necessity. The concrete implementation choice – Laminar deployed locally in Docker and combined with MLflow for prompt and schema versioning – should be described in detail in Chapter 3 as part of the system architecture and demonstrated in Chapter 4 through traces, dashboards, and evaluation results rather than fully expanded here.

2.5 Synthesis and Gap Analysis for the VIP Project

2.5.1 Identified Gaps in Existing Solutions

The state of the art confirms that the ecosystem is already mature, but in fragmented ways. n8n provides strong workflow composition, integrations, scalable workers, and observability hooks, yet it remains primarily centered on workflow automation. CrewAI provides a better abstraction for collaborative agents, flows, state, and provider flexibility, yet it remains a framework that still needs surrounding platform design choices for deployment lifecycle, standardization, and governance. Managed-product platforms such as Jarvio highlight the operational value of ready-to-use agent experiences, but they do not provide the same architectural transparency and portability needed for a configurable enterprise AI hub. The missing combination is therefore clear: a platform that can generate agents from declarative specifications, package them as reusable deployment units, orchestrate them visually and programmatically, observe them deeply in runtime, and choose providers through continuous evaluation rather than vendor habit.

2.5.2 Design Implications for VIP

These gaps translate directly into design implications for VIP. First, agent creation must be configuration-driven rather than hard-coded, so that prompts, schemas, tools, and providers can be modified without rewriting application logic. Second, the provider layer must remain abstract and replaceable, because model choice is a moving engineering decision shaped by quality, latency, privacy, and cost. Third, observability must be embedded by design through traces, metrics, and execution metadata rather than added after deployment. Fourth, prompts and schemas must be versioned as governed artifacts, because reproducibility and rollback are essential in enterprise environments. Finally, model selection must be connected to a dedicated evaluation loop, which is precisely the role of the second project, SLM-Evals. VIP is therefore positioned not as another isolated framework, but as an integrating architecture that combines generation, orchestration, observability, and evaluation in one platform-level design.

2.6 Preliminary Study of the SLM-Evals Second Project

Alongside the main VIP platform, this internship includes a second project named SLM-Evals, dedicated to the systematic evaluation of language-model providers and candidate models before their integration into production workflows. In the current VIP draft, SLM-Evals is already introduced as a reproducible benchmark pipeline comparing providers under controlled prompts and golden datasets while tracking latency, token usage, and layered evaluation metrics. This subsection formalizes that role by defining the motivation, the evaluation scope, the main measurement dimensions, and the contribution of this second project to VIP.

2.6.1 Motivation of the SLM-Evals Second Project

The motivation behind SLM-Evals is that model selection in enterprise agentic systems cannot rely on vendor marketing claims or isolated public benchmark results. VIP is intended to orchestrate configurable agents across heterogeneous business workflows, which means that the selected model must not only be “strong” in a general sense, but also sufficiently reliable for structured outputs, sufficiently fast for multi-step orchestration, and sufficiently economical for repeated production use. HELM explicitly argues for multi-dimensional evaluation rather than single-score benchmarking, and this principle is particularly important for agentic systems where quality, efficiency, and trustworthiness must be optimized together.

In this report, the distinction between LLMs and SLMs is operational rather than purely semantic. LLMs refer to frontier or larger-capacity models, including proprietary APIs and large open-weight families, which are typically used as high-quality baselines for reasoning-intensive tasks. SLMs refer to more compact models that are easier to deploy, cheaper to run, and better suited to memory- or latency-constrained environments. Microsoft’s Phi-4 model card explicitly positions Phi-4 as a 14B dense model intended for memory/compute-constrained and latency-bound scenarios, while Meta’s Llama 3.1 family spans 8B, 70B, and 405B variants with 128k context, making it useful as an open-weight comparison family across scales. Both classes must therefore be evaluated, because VIP may need a higher-capacity model for difficult reasoning steps and a smaller or locally deployable model for routing, extraction, or budget-

sensitive tasks.

From this perspective, the problem addressed by SLM-Evals can be stated as follows: how can heterogeneous providers and models be compared under one fair, repeatable protocol so that VIP can select the most appropriate model for each agent task while controlling cost, latency, and reliability risk? This problem is directly aligned with the current VIP draft, which already identifies provider flexibility, reduced vendor lock-in, and evidence-based provider selection as explicit project goals.

2.6.2 Provider and Model Evaluation Need

The provider/model evaluation need emerges from two complementary decisions. The first is a provider-level decision: which execution environment or API vendor should be preferred under given cost, latency, and governance constraints. The second is a model-level decision: which exact model should be assigned to a given agent or task category. These two decisions are related but not identical. For example, OpenAI exposes tiered pricing across flagship and smaller models, Groq emphasizes high token speed and publishes model-dependent throughput and per-token prices, while Google positions Gemini Flash as a price-performance option for low-latency, high-volume tasks. Such differences make provider choice an operational decision, not just a research preference.

At the same time, model-level choice matters because open-weight and compact models are now strong enough to be viable for some enterprise workloads, especially when the task is narrow, schema-constrained, or repeated at scale. Phi-4 is explicitly described as a high-quality compact model for constrained environments, while Llama 3.1 provides a scalable open-weight family that can be tested across small and large variants. This is precisely why SLM-Evals must evaluate both LLMs and SLMs: not to identify a universal winner, but to determine which model is "good enough" for a given task type, under a measurable latency and cost envelope.

From the VIP perspective, the evaluation need is therefore explicit. A viable provider/model pair must satisfy at least five conditions: acceptable task quality, predictable latency, affordable token cost, reliable structured-output behavior, and sufficient operational traceability for debugging and governance. Because VIP agents exchange schema-constrained data and execute inside orchestrated workflows, schema adherence and execution observability are not secondary concerns; they are fundamental selection criteria. This justifies the existence of SLM-Evals as a dedicated experimental layer rather than an ad hoc benchmark notebook.

2.6.3 Main Evaluation Dimensions

At the time of writing, the exact final benchmark bundle, evaluation budget, and final provider list remain partially unspecified. A practical default protocol is therefore to combine one broad knowledge/reasoning benchmark, one arithmetic reasoning benchmark, one factuality benchmark, and one VIP-specific golden dataset reflecting the project's own agent tasks. Reasonable default public datasets are MMLU or MMLU-Pro for broad multi-domain reasoning, GSM8K for multi-step arithmetic reasoning, and TruthfulQA for factual reliability. If code-generation or code-transformation agents later become part of scope, HumanEval can be added as an optional benchmark. The golden dataset remains the most important component

for VIP, because public benchmarks do not directly measure schema-heavy enterprise tasks, system-specific prompts, or business-specific failure modes.

The evaluation dimensions should cover both output quality and operational behavior. For closed-form tasks, the primary metrics are accuracy, exact match, or task-specific correctness. For open-ended outputs, a rubric-based LLM-as-a-judge score can be used, provided that the judge model and evaluation prompt remain fixed across all compared candidates. For schema-constrained tasks, structured-output correctness must be measured explicitly through JSON validity, schema-validation pass rate, and field-level exact match. This requirement is technically justified because OpenAI, Groq, and Gemini all expose schema-constrained structured-output mechanisms, but with different guarantees and model coverage. Groq, for instance, distinguishes strict mode from best-effort mode and explicitly documents exact schema adherence only for supported models; OpenAI recommends Structured Outputs over plain JSON mode because schema adherence is stronger than mere JSON validity; and Gemini documents JSON-Schema-constrained responses for structured extraction and agentic workflows.

In addition to output quality, SLM-Evals must track latency, token usage, estimated cost, and reliability. Latency should be reported at least through p50 and p95 response time. Token usage should distinguish prompt and completion tokens whenever provider metadata allows it. Estimated cost should be derived from provider pricing tables and logged per request and per evaluation batch. Reliability should include timeout rate, API error rate, retry rate, and parse-failure rate. Since factual reliability is particularly important in knowledge-intensive workflows, a separate hallucination or unsupported-claim rate should be measured either with reference-grounded scoring or with truthfulness-oriented benchmarks such as TruthfulQA. Finally, if the model is prompted to emit confidence values, calibration can be evaluated through measures such as Expected Calibration Error or Brier score, which is consistent with HELM's inclusion of calibration as a first-class evaluation dimension and with recent work on confidence calibration in LLMs.

The experimental protocol must remain reproducible. All compared runs should use the same prompt template, the same judge prompt if LLM-as-a-judge is used, the same input schema, the same output schema, and the same maximum-token budget. Prompt versions and schemas should be versioned in MLflow, and the evaluation runner should be executed in Docker to stabilize dependencies and environment differences. When a provider supports deterministic controls, a fixed seed and low-temperature setting should be used; otherwise, deterministic settings such as temperature zero and repeated runs should be preferred, with mean and variance reported for stochastic conditions. MLflow now provides provider-agnostic evaluation, datasets, scorers, and tracing support for LLM applications, while Laminar provides dataset-driven evaluations, automatic tracing of model invocations, and storage of per-datapoint results and traces over time. These capabilities make both reproducibility and later auditability much easier to enforce.

2.6.4 Contribution of the Second Project to VIP

The contribution of SLM-Evals to VIP is not limited to producing a benchmark table. Its real value lies in transforming evaluation results into provider-selection knowledge that can be

consumed by the VIP platform. In other words, SLM-Evals should feed VIP with a structured decision layer: which provider/model pair is the default for a given agent family, which fallback should be triggered when cost or latency thresholds are exceeded, and which task categories require escalation to a stronger frontier model. This logic is already consistent with the current VIP draft, which states that SLM-Evals supports provider flexibility and guides the provider layer.

Concretely, the outputs of SLM-Evals can be reused by VIP in three ways. First, they can define default deployment choices for agent archetypes such as extraction agents, routing agents, or reasoning agents. Second, they can define runtime policies, for example: use the cheapest model that still satisfies a schema-pass threshold for structured extraction; escalate to a stronger model when confidence is low or when the task is reasoning-heavy; or prefer a local/open-weight model when privacy or budget constraints dominate. Third, they can support a continuous improvement loop in which production traces are collected through Laminar, difficult or failing examples are promoted into evaluation datasets, and the evaluation suite is re-run before updating the VIP routing policy. Laminar explicitly supports trace collection, dataset building, replay, and evaluation runs over datasets, while MLflow supports prompt versioning, provider-agnostic evaluation, and centralized experiment tracking.

For this reason, SLM-Evals should be presented in Chapter 2 as a preliminary study and evaluation foundation, not as an isolated experiment. The detailed integration architecture can then be deferred to Chapter 3, while the concrete execution protocol, obtained results, and provider-selection outcomes can be reported in Chapter 4. This separation keeps Chapter 2 analytical and preparatory, while preserving implementation detail for the later chapters.

2.7 Chapter Summary